END-SEMESTER PROJECT · NOVEMBER 2026

ApexHunter

Finding driver mistakes and pit-stop windows in Formula 1 telemetry, and an honest test of how well it works

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92	57.9 M	2
sessions	telemetry rows	seasons

THE PROBLEM

Two questions asked of the car's data

1 Where did the driver make a mistake?

Moments where he did something different from the fastest drivers at the same spot, in a way that cost time: a lock-up, an early lift, a missed downshift.

2 When should he change tyres?

From the laps driven so far, a five-lap pit window, and a "pit now" call when a safety car makes the stop cheaper. Checked against the stops drivers really made.

The catch: there is no answer key.
Nobody labels mistakes, and a team's pit lap is a decision, not a correct answer. So every model is scored on tests we can check (a controlled test on synthetic mistakes, races it never saw) and is compared with a simple baseline.

THE PROBLEM

Why this is hard

0.05 s

lost per lap of tyre age, on average

0.9 s

normal spread of lap time from one lap to the next (traffic, wind, small errors)

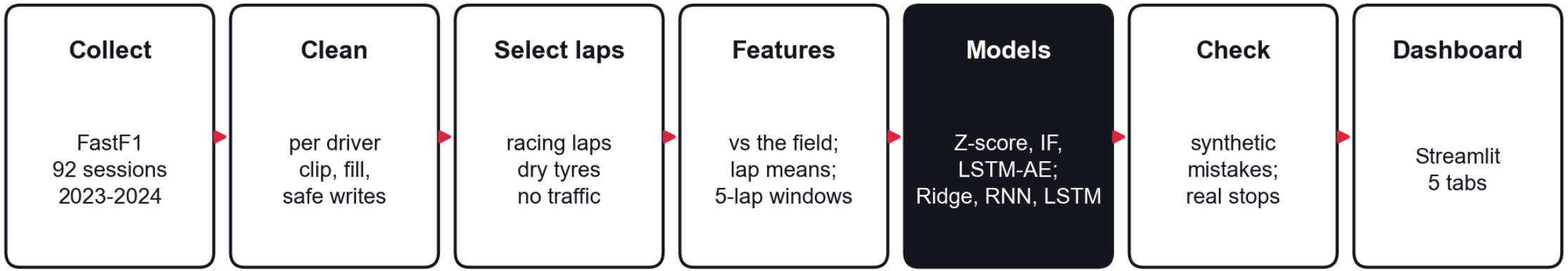
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labelled mistakes in any public F1 data

So the aim is not a perfect strategist.
It is to find what the data really supports, and prove it is not luck, a data leak or overfitting.

WHAT WE BUILT

A pipeline from raw telemetry to a dashboard



Simple baseline

Z-score rule · "no change" (next lap = last lap)

Machine learning

Isolation Forest · Ridge regression

Deep learning

LSTM autoencoder · Vanilla RNN · LSTM

DATA

Two seasons of Formula 1 telemetry

92

sessions: every race and qualifying of 2023 and 2024

57.9 M

telemetry rows: speed, throttle, brake, gear, RPM, position

~50,000

race laps with stint, tyre compound, tyre age and pit times

4 / s

samples per second, irregularly spaced (0.014–0.76 s)

Also used: gap to the car ahead and DRS state (to leave traffic out), corner positions (to name where an event happened), track status (safety car, VSC, red flag).

DATA QUALITY

Fixing the data came first

Problem we found	What it would have caused	Fix
Detector compared absolute values	95 % of "mistakes" were ordinary braking	Deviation from the field at the same spot
Pit, safety-car and first laps kept	Slow laps read as mistakes and tyre wear	Racing laps only (one shared filter)
Wet tyres mixed with dry (2,460 laps)	Rain laps read as extreme wear	Dry tyres with known tyre age
Slipstream and dirty air kept	Overtakes flagged as mistakes	Leave out DRS-open or < 1 s behind
Random train/test split of windows	Near-identical windows on both sides	Split by whole races, latest for test
Scaler fitted on all data	Test statistics leaked into training	Scalers fitted on training races only

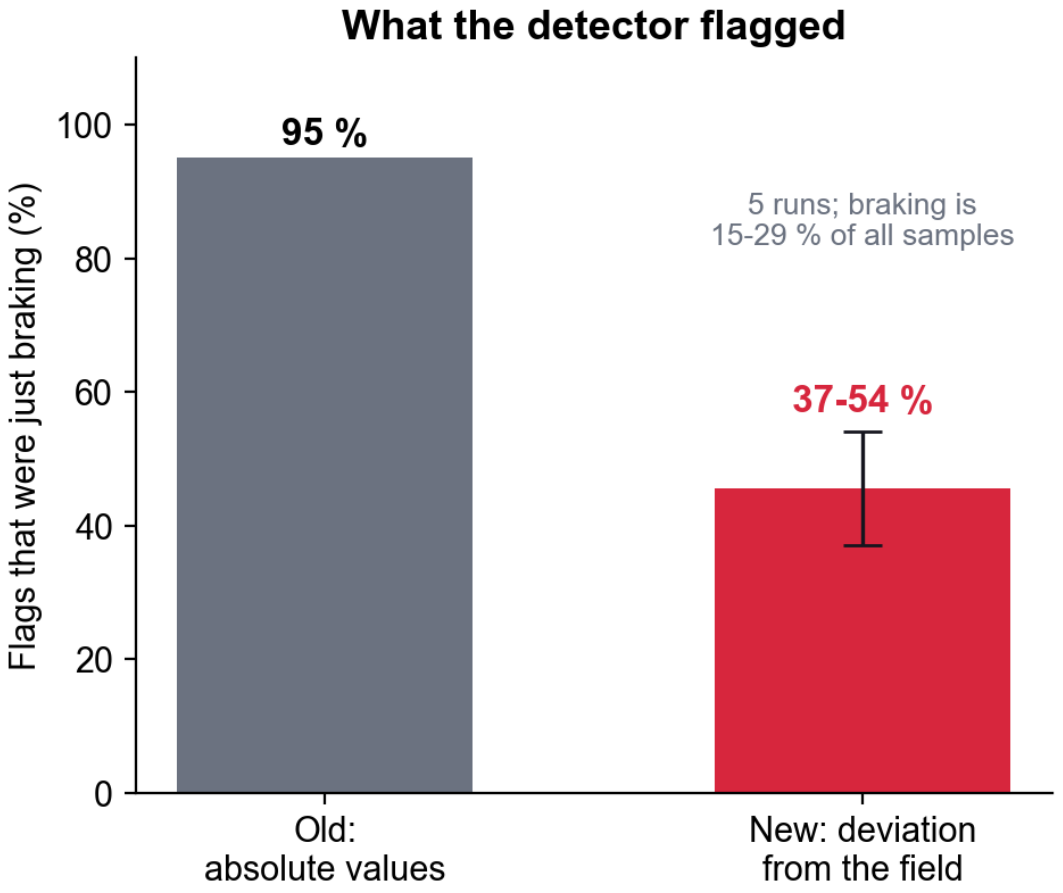
55 problems logged and fixed in total, each with a test; 197 tests, none faking data or models

Compare with the field, not with nothing

Old question: "Is this sample unusual?" Braking is rare, so braking looked like a mistake.

New question: "Is the driver doing something unusual compared with the fastest drivers at this exact point on track?"

Five features, each a deviation from the field per 10 m of track, divided by the field's spread there: speed, throttle, brake, gear, acceleration.



QUESTION 1 · MISTAKES

Three detectors, two checks

Z-score (baseline)

Flags a sample when any one feature is extreme.

Isolation Forest

Random trees isolate unusual points in few splits; 200 trees, 2 % flagged.

LSTM autoencoder

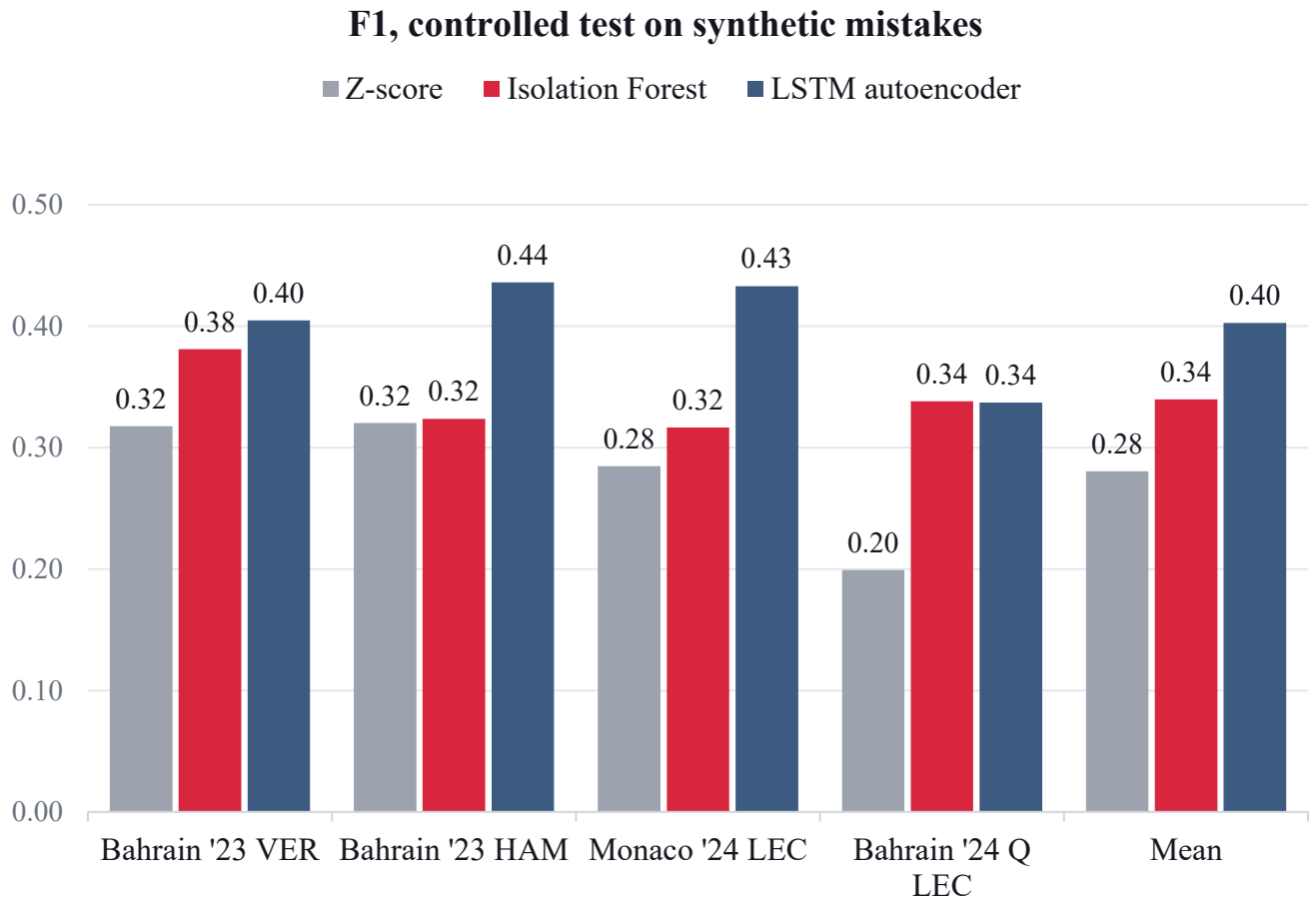
Rebuilds ~2 s of driving; what it cannot rebuild is unusual. 2 % flagged.

Synthetic mistake	What is changed
Lock-up	extra 15–30 km/h drop in a braking zone
Early lift	throttle off 0.5–1 s early
Late throttle	throttle held at 0 for 0.5–1 s
Pedal overlap	throttle 20–40 % while braking
Missed downshift	one gear too high in a slow corner

1. Controlled test on synthetic mistakes: precision, recall, F1. Its limit: it shows these patterns are found, not that every real flag is a mistake.
2. Real-lap check: do flagged moments cost time against the driver's own laps, more often than random places?

QUESTION 1 · RESULTS

Both learned detectors beat the baseline



0.34

Isolation Forest mean F1 (Z-score 0.28)

0.40

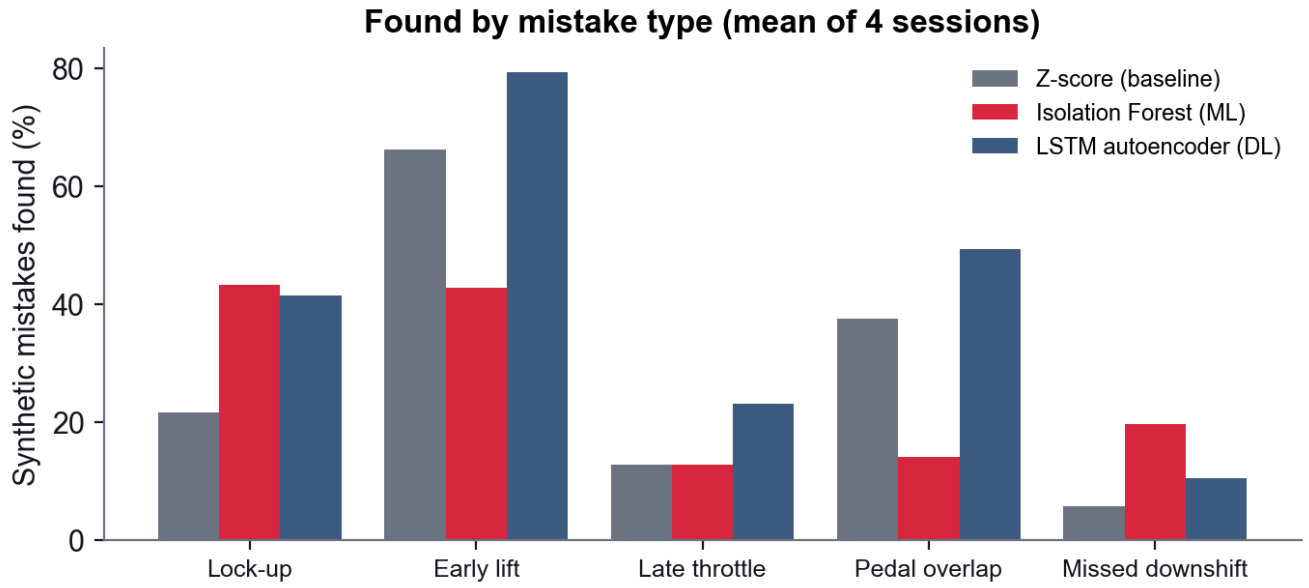
LSTM autoencoder mean F1: finds the most synthetic mistakes

18 %

of Isolation Forest flags cost time on real laps (random places 12 %, autoencoder 14 %)

QUESTION 1 · RESULTS

Each detector is good at different mistakes



Top events, Hamilton, Bahrain 2023 (IF)

1. **Turn 10, lap 38**
   +0.64 s · Accelerating harder than the field
2. **Approaching Turn 12, lap 34**
   +0.52 s · 24 km/h slower than the field here
3. **Turn 4, lap 37**
   +0.42 s · 14 km/h slower than the field here
4. **Turn 4, lap 47**
   +0.39 s · 19 km/h slower than the field here

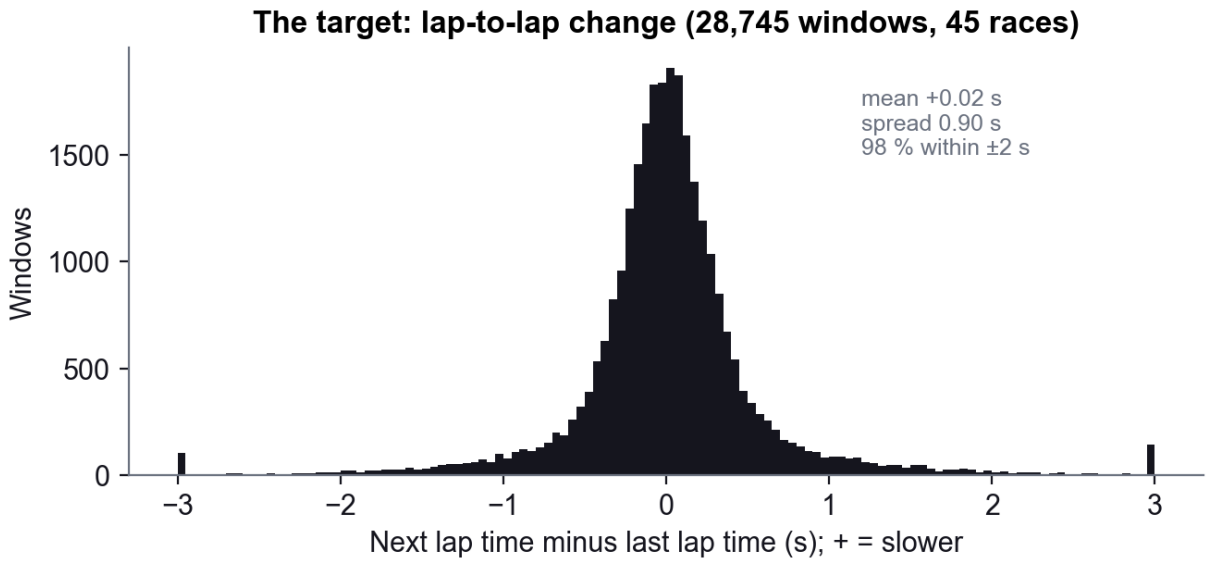
QUESTION 2 · TYRES

Predict how much slower the next lap will be

The model sees: the driver's last 5 laps — how fast each was, how he used throttle and brake, how old the tyres are and which compound.

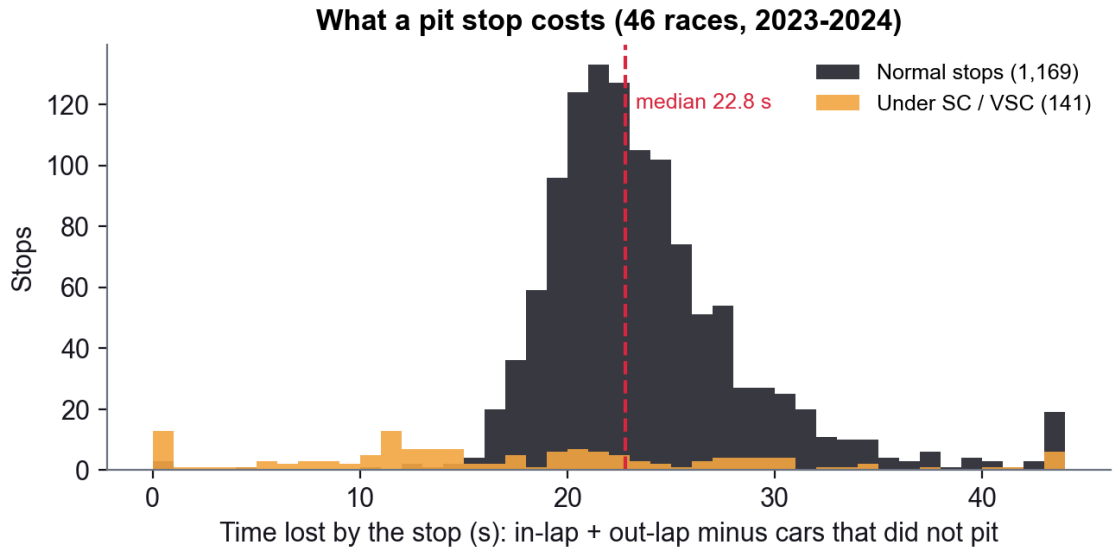
It predicts: how many seconds slower (or faster) the next lap will be.

Why the change, not the lap time: a lap time mostly tells you which track it is. The change is what worn tyres actually affect.



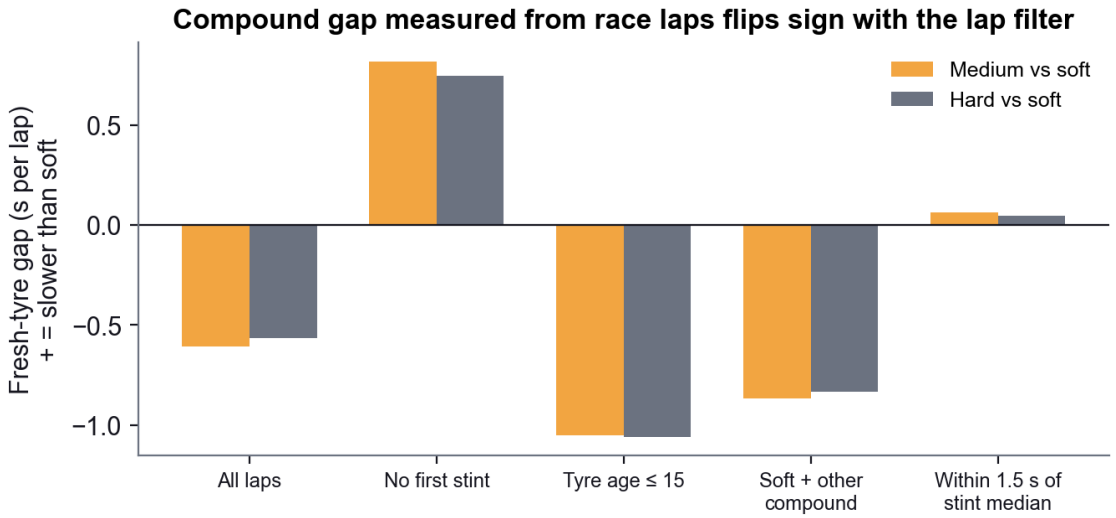
Most laps are within ±1 s of the one before. Simply repeating the last lap ("no change") is off by 0.39 s on average: the baseline every model is compared with.

What a stop costs, and what the data cannot tell



22.8 s lost by a normal pit stop (median of 46 races)

× 0.7 cheaper under a safety car: everyone is slow anyway

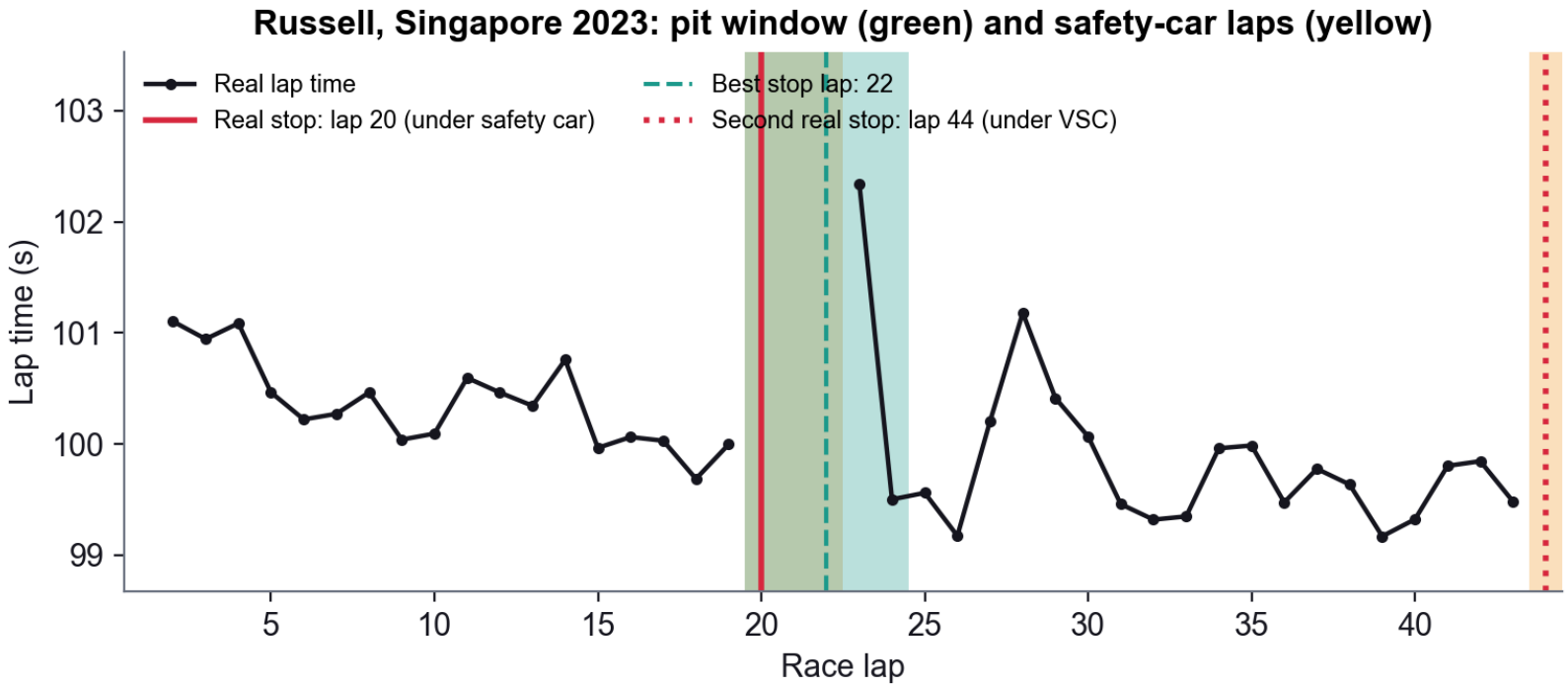


Soft, medium and hard look almost the same in race data (the measured gap even changes sign). So we do not advise a compound; the strategy uses the tyre the driver actually fitted.

QUESTION 2 · TYRES

From predicted wear to a five-lap pit window

1 Estimate tyre wear per lap. **2** For each possible stop lap, add tyre time lost + stop cost. **3** **Cheapest lap ± 2 laps = the window.**



Russell, Singapore 2023

Window laps 20–24, best lap 22. Safety car on lap 20: the stop is 30 % cheaper, so "pit now".

He really pitted on lap 20, inside the window.

DEEP LEARNING

Recurrent networks

An LSTM autoencoder for mistakes; vanilla RNN vs LSTM for tyres. Same data, same tests, same baselines as the other models

An LSTM autoencoder learns normal driving

1 Input

16 readings (~2 s) of the five deviation features

2 Encoder → decoder

an LSTM squeezes the window into 16 numbers, a second LSTM rebuilds it

3 Score

rebuild error; the 2 % worst readings are flagged, as for the Isolation Forest

0.40

mean F1 on synthetic mistakes, the best of the three (Isolation Forest 0.34)

0.34

recall: it finds most mistakes that unfold over time, such as early lifts

14 %

of its real flags cost time: about as often as random places (12 %)

DEEP LEARNING

Two recurrent networks read the five laps in order

1 Vanilla RNN

Reads one lap at a time and keeps a running memory of the laps so far. Over long sequences that memory fades (vanishing gradients).

2 LSTM

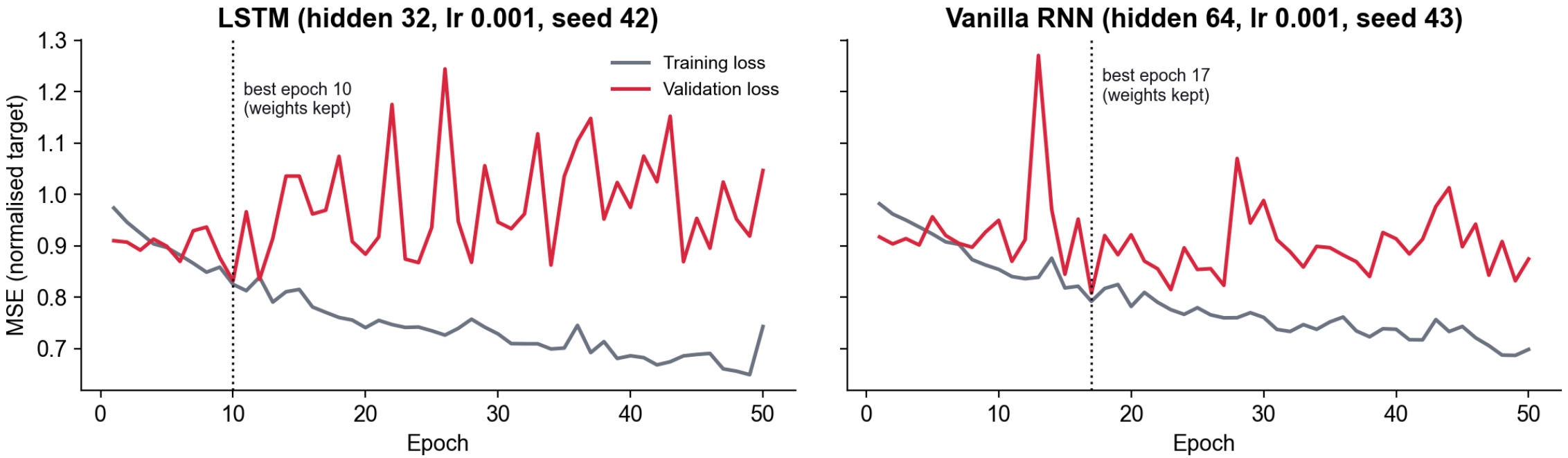
Adds gates that decide what to remember, what to forget and what to pass on, so the memory does not fade.

3 Same setup for both

Same laps, same test races, same baseline. Only the memory cell differs, so the comparison is fair.

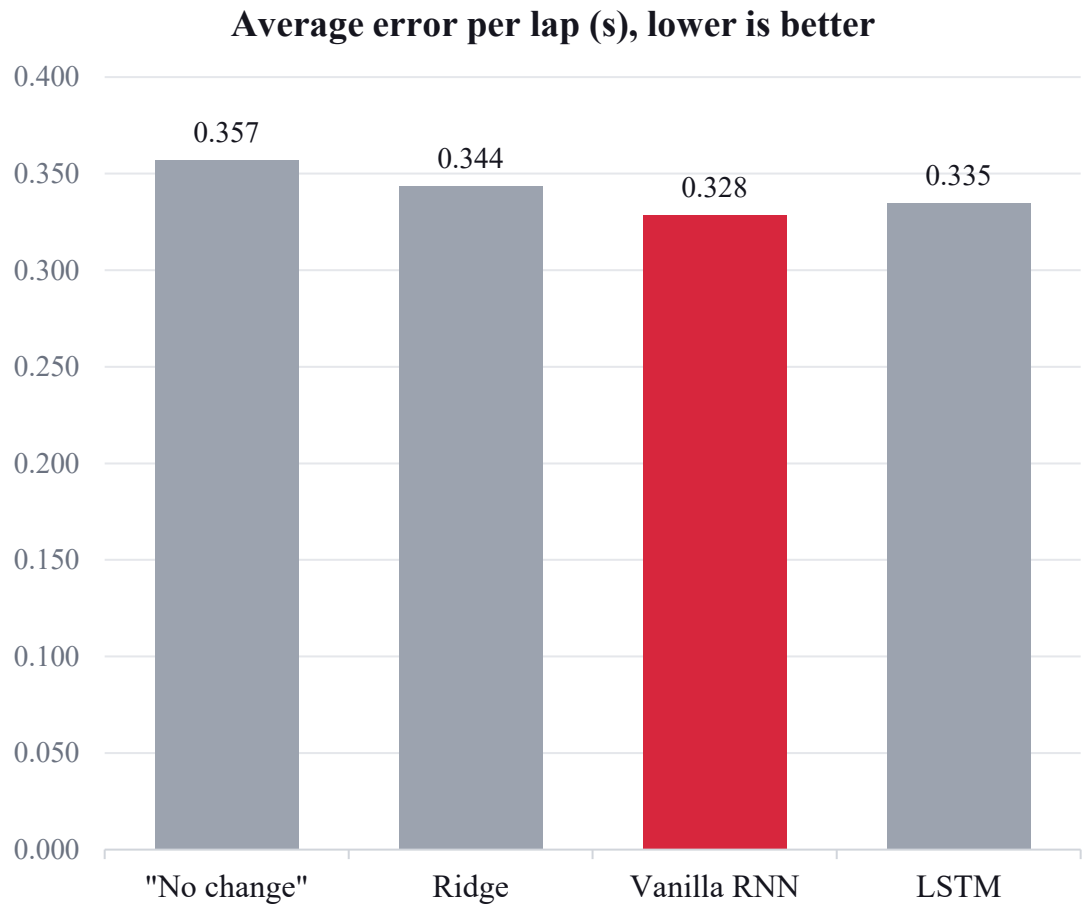
How they were trained: settings (size, learning rate) chosen on the most recent training races; the best epoch is kept; each network trained 3 times with different random starts and the average reported. Test races are used once.

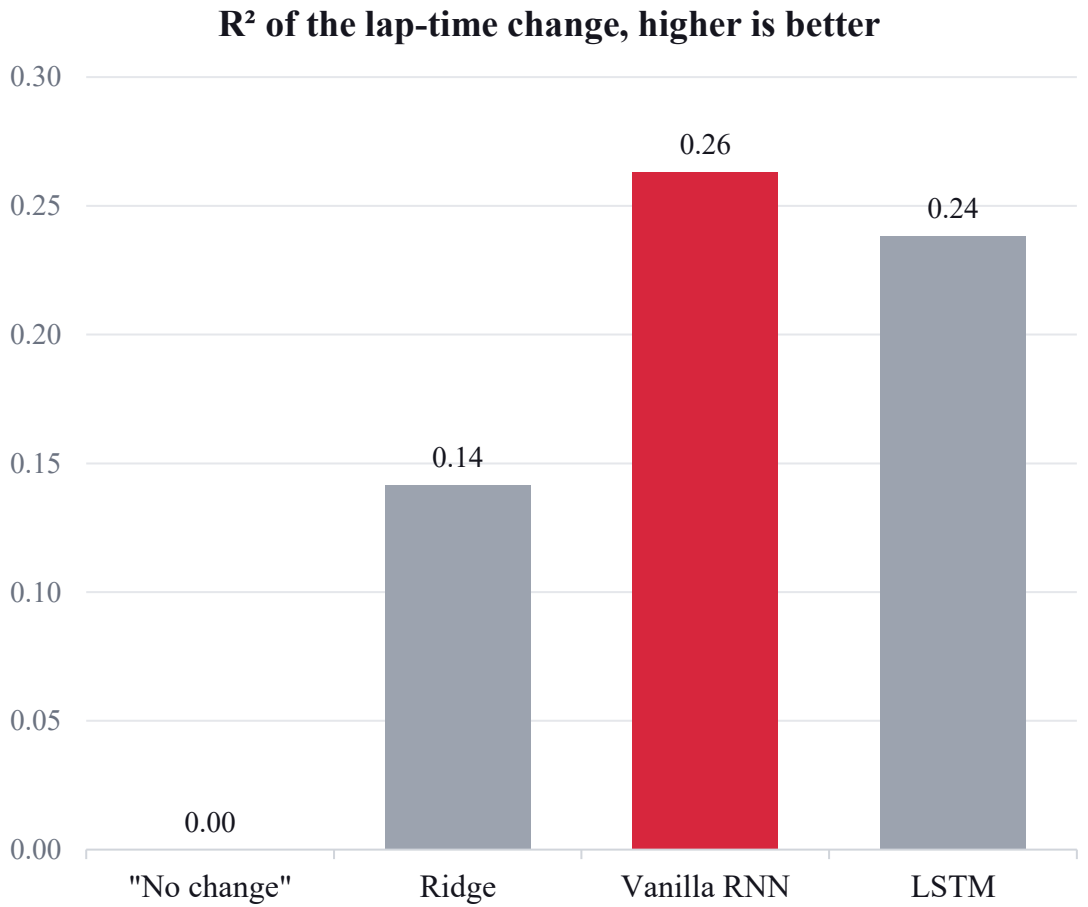
After 10–17 epochs the networks start to memorise



Training error keeps falling; validation error stops improving and becomes noisy. Most lap-to-lap change is noise the inputs cannot explain, so extra training only memorises it. The weights of the best validation epoch are restored.

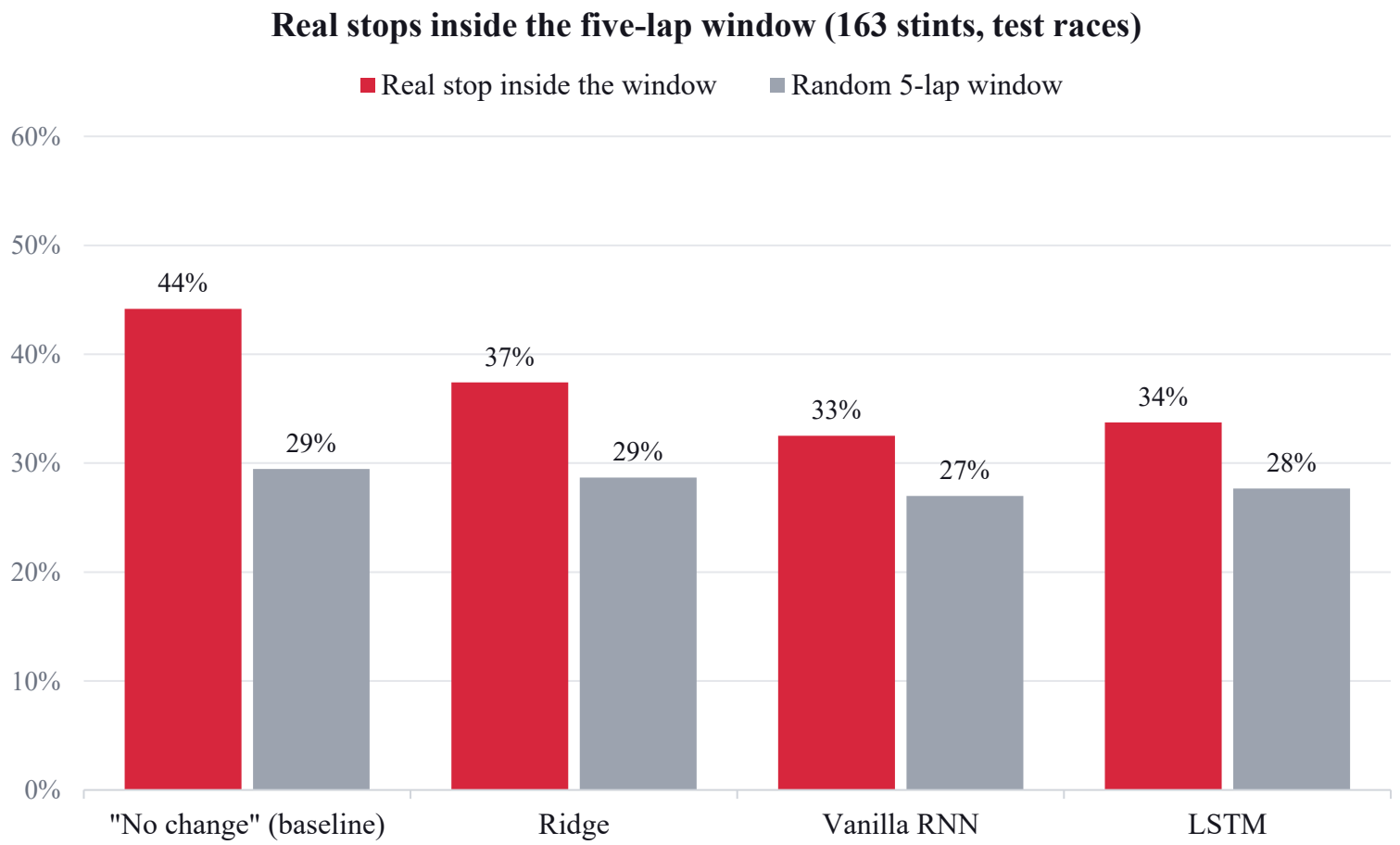
Both networks beat Ridge on lap times





Saved models shown. Mean over 3 seeds: RNN 0.332 ± 0.007 s, LSTM 0.338 ± 0.004 s; with 5-lap sequences the LSTM's gates bring no advantage

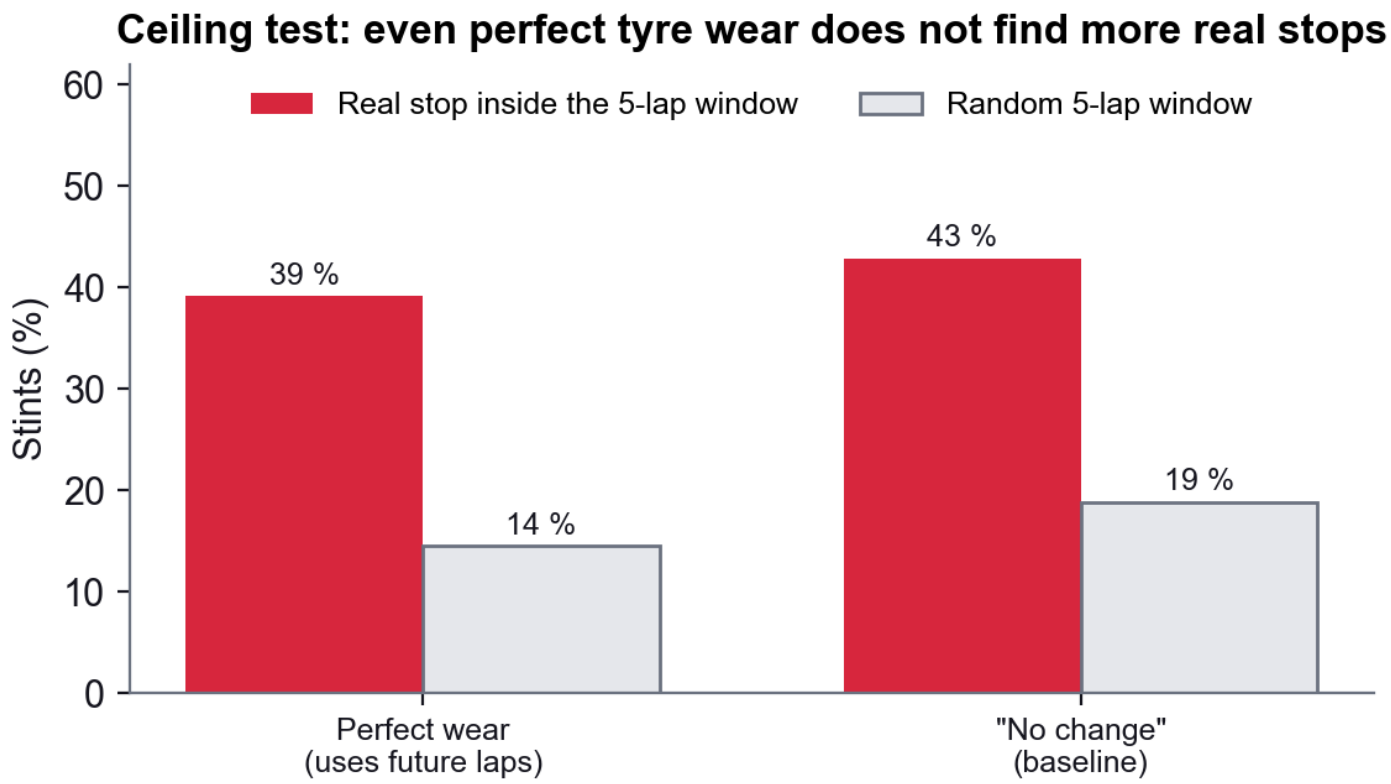
Better lap times do not mean better pit windows



Every model beats chance
the real stop is inside the window more often than in a random one.

The trained models do not beat the baseline.
Their predictions correct lap-to-lap noise rather than measure wear; averaged over a stint they often predict laps getting faster, which gives no window.

Even perfect tyre wear would not find more real stops



Why?

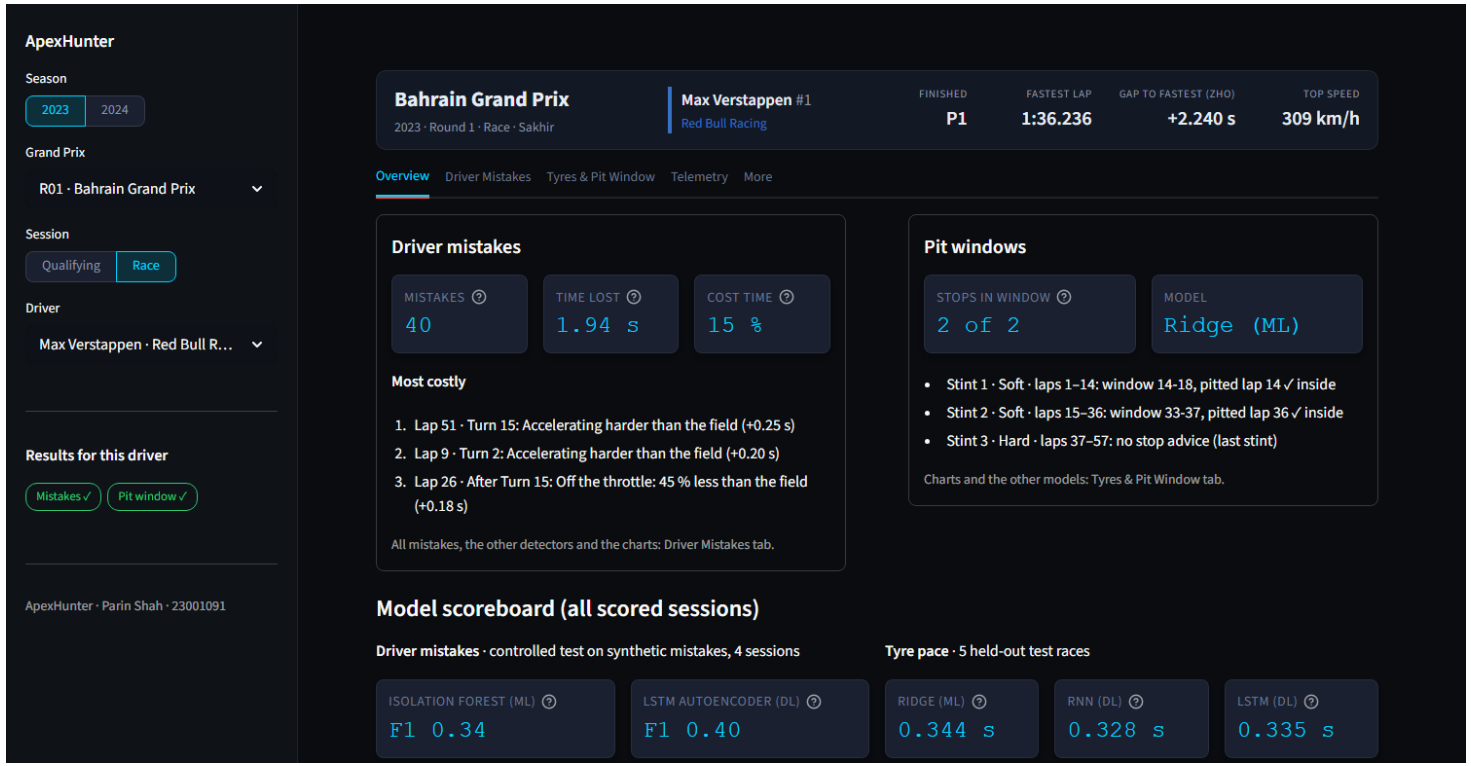
Real stops are set by undercuts, traffic, team plans and safety cars.

Wear is so small (0.05 s/lap) that the best stop lap barely moves.

Published work agrees: on F1 tyre data a tree model beat RNN, LSTM, GRU and a transformer (Todd et al., 2025).

THE DASHBOARD

Five tabs, redesigned around the two questions



Overview

both answers and the model scoreboard

Driver Mistakes

three detectors, map, list, each mistake against the field

Tyres & Pit Window

Ridge / RNN / LSTM, window vs real stop

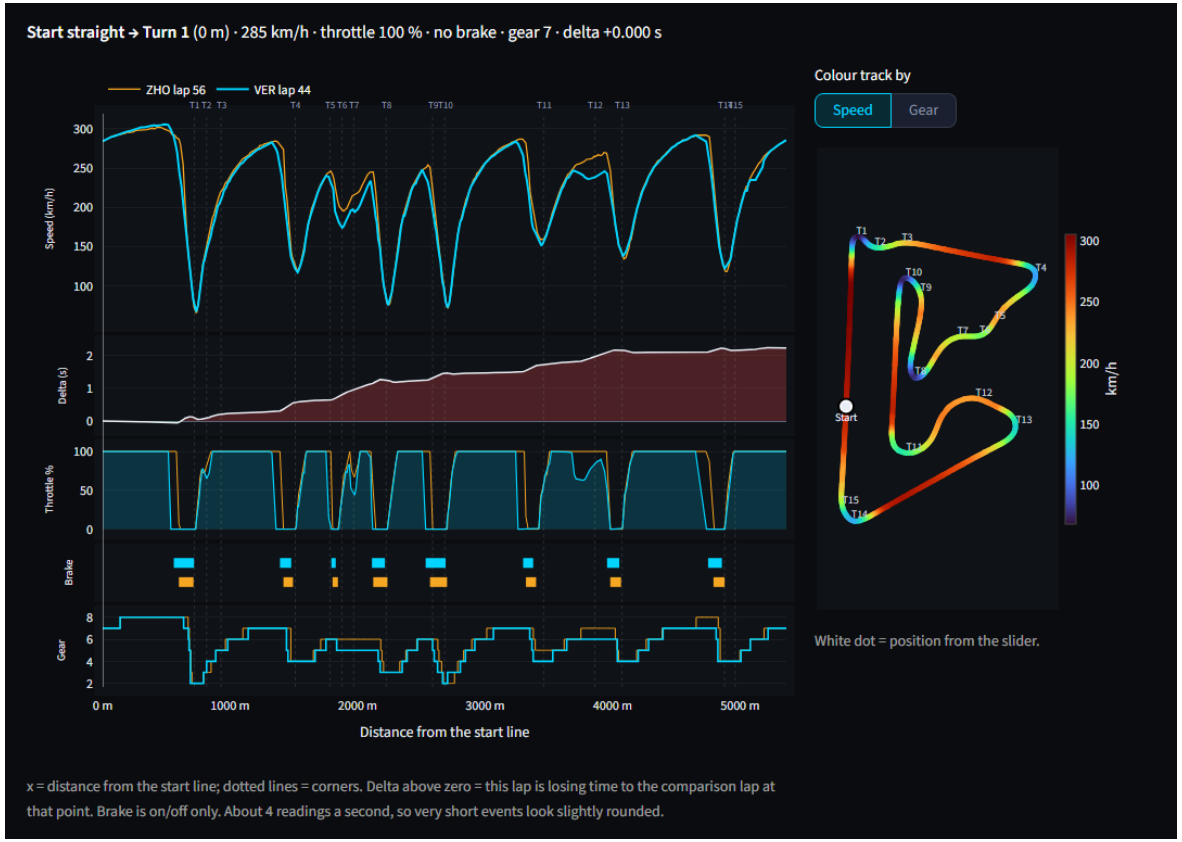
Telemetry

any lap of any driver, against distance

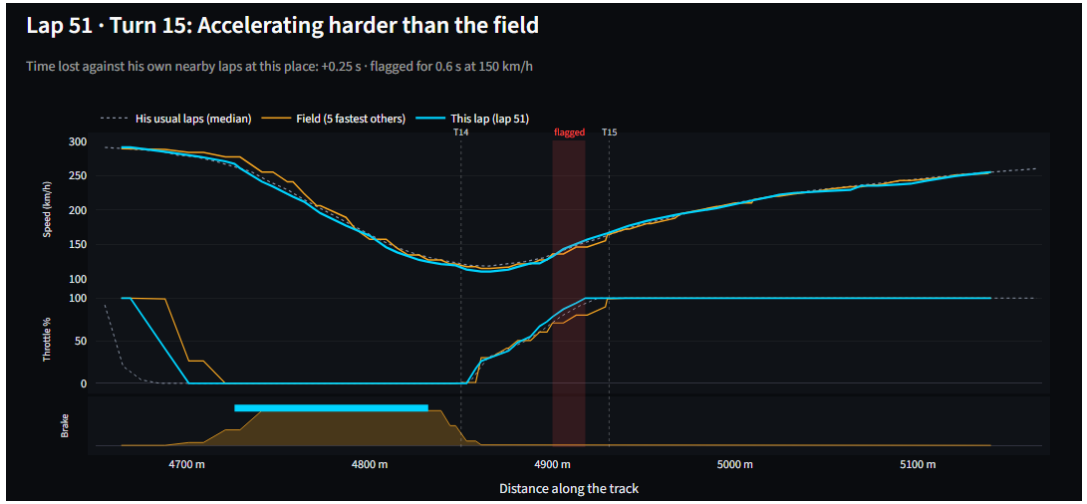
More

apex video, racing line, Big Data

Each chart answers one question



Telemetry against distance: speed, time delta, throttle, brake bands, gear, track map



Why a mistake was flagged: this lap (blue) against the field (orange) and his usual laps (dotted), with the flagged stretch in red.

197 automated tests · 55 bugs fixed · telemetry for all 92 sessions

WHAT WE LEARNED

Six lessons from the project

01 Clean the data first

55 problems would have made later numbers wrong.

02 Compare with the right reference

The field at the same spot turned braking flags into mistakes.

03 Build a test with an answer key

Synthetic mistakes made an unlabelled problem measurable; a real-lap check keeps it honest.

04 Keep a simple baseline

"No change" beat every model on pit windows.

05 Deep learning is not automatic

The autoencoder wins the synthetic test but not on real laps; the plain RNN ties the LSTM.

06 Report what doesn't work

The ceiling test is the project's most useful result.

LIMITS AND NEXT STEPS

What we can't claim yet

Limitations

Synthetic mistakes show the detector finds these patterns, not that every flag is a real mistake.

Only four sessions scored for mistakes; five test races for tyres.

A tyres-only strategy: traffic, rivals and track position are left out on purpose.

Next steps

Predict the laps until a driver pits directly.

Add traffic and rival cars to the strategy.

Mistake and pit-window outputs for every driver of every race.

APEXHUNTER

Thank you

Questions are welcome

Mistakes: Isolation Forest F1 0.34, LSTM autoencoder 0.40 vs Z-score 0.28 · Tyres: RNN and LSTM best on lap times; pit window above chance